Q1 - 2024 (04 Apr Shift 1)

The vertices of a triangle are $\mathrm{A}(-1,3), \mathrm{B}(-2,2)$ and $\mathrm{C}(3,-1)$. A new triangle is formed by shifting the sides of the triangle by one unit inwards. Then the equation of the side of the new triangle nearest to origin is :

(1) $x+y+(2-\sqrt{2})=0$

(2) $-x+y-(2-\sqrt{2})=0$

(3) $x+y-(2-\sqrt{2})=0$

(4) $x-y-(2+\sqrt{2})=0$

Q2 - 2024 (04 Apr Shift 2)

Consider a triangle ABC having the vertices $\mathrm{A}(1,2), \mathrm{B}(\alpha, \beta)$ and $\mathrm{C}(\gamma, \delta)$ and angles $\angle ABC=\frac{\pi}{6}$ and $\angle BAC=\frac{2\pi}{3}$. If the points B and C lie on the line $y=x+4$, then $\alpha^2+\gamma^2$ is equal to ________

Q3 - 2024 (05 Apr Shift 1)

Let two straight lines drawn from the origin O intersect the line $3x+4y=12$ at the points P and Q such that $\triangle\mathrm{OPQ}$ is an isosceles triangle and $\angle\mathrm{POQ}=90^{\circ}$. If $l=\mathrm{OP}^2+\mathrm{PQ}^2+\mathrm{QO}^2$, then the greatest integer less than or equal to l is :

(1) 42

(2) 46

(3) 44

(4) 48

Q4 - 2024 (05 Apr Shift 1)

If $\mathrm{A}(1,-1,2), \mathrm{B}(5,7,-6), \mathrm{C}(3,4,-10)$ and $\mathrm{D}(-1,-4,-2)$ are the vertices of a quadrilateral $ABCD$, then its area is :

(1) $48\sqrt{7}$

(2) $12\sqrt{29}$

(3) $24\sqrt{7}$

(4) $24\sqrt{29}$

Q5 - 2024 (05 Apr Shift 2)

Let $A(-1,1)$ and $B(2,3)$ be two points and P be a variable point above the line AB such that the area of $\triangle \mathrm{PAB}$ is 10 . If the locus of P is $\mathrm{a}x+\mathrm{b}y=15$, then $5\mathrm{a}+2\mathrm{~b}$ is :

(1) 6

(2) $-\frac{6}{5}$

(3) 4

(4) $-\frac{12}{5}$

Q6 - 2024 (06 Apr Shift 1)

Let a variable line of slope $m>0$ passing through the point $(4,-9)$ intersect the coordinate axes at the points A and B. The minimum value of the sum of the distances of A and B from the origin is

(1) 30

(2) 25

(3) 15

(4) 10

Q7 - 2024 (06 Apr Shift 2)

If $\mathrm{P}(6,1)$ be the orthocentre of the triangle whose vertices are $\mathrm{A}(5,-2), \mathrm{B}(8,3)$ and $\mathrm{C}(\mathrm{h}, \mathrm{k})$, then the point C lies on the circle:

(1) $x^2+y^2-61=0$

(2) $x^2+y^2-52=0$

(3) $x^2+y^2-65=0$

(4) $x^2+y^2-74=0$

Q8 - 2024 (06 Apr Shift 2)

If the locus of the point, whose distances from the point $(2,1)$ and $(1,3)$ are in the ratio $5:4$, is $ax^2+by^2+cxy+dx+ey+170=0$, then the value of $a^2+2b+3c+4d+e$ is equal to :

(1) 37

(2) 437

(3) -27

(4) 5

Q9 - 2024 (08 Apr Shift 1)

The equations of two sides AB and AC of a triangle ABC are $4x+y=14$ and $3x-2y=5$, respectively. The point $\left(2,-\frac{4}{3}\right)$ divides the third side BC internally in the ratio $2:1$. the equation of the side BC is

(1) $x+3y+2=0$

(2) $x-6y-10=0$

(3) $x-3y-6=0$

(4) $x+6y+6=0$

Q10 - 2024 (08 Apr Shift 1)

If the orthocentre of the triangle formed by the lines $2x+3y-1=0$, $x+2y-1=0$ and $ax+by-1=0$, is the centroid of another triangle, whose circumcentre and orthocentre respectively are $(3,4)$ and $(-6,-8)$, then the value of $|a-b|$ is_______

Q11 - 2024 (08 Apr Shift 2)

If the image of the point $(-4,5)$ in the line $x+2y=2$ lies on the circle $(x+4)^2+(y-3)^2=r^2$, then r is equal to:

(1) 2

(2) 3

(3) 1

(4) 4

Q12 - 2024 (08 Apr Shift 2)

If the line segment joining the points $(5,2)$ and $(2,a)$ subtends an angle $\frac{\pi}{4}$ at the origin, then the absolute value of the product of all possible values of a is :

(1) 6

(2) 8

(3) 2

(4) -4

Q13 - 2024 (08 Apr Shift 2)

Let a ray of light passing through the point $(3,10)$ reflects on the line $2x+y=6$ and the reflected ray passes through the point $(7,2)$. If the equation of the incident ray is $ax+by+1=0$, then a^2+b^2+3ab is equal to________

Q14 - 2024 (09 Apr Shift 1)

A ray of light coming from the point $P(1,2)$ gets reflected from the point Q on the x-axis and then passes through the point $R(4,3)$. If the point $S(h,k)$ is such that PQRS is a parallelogram, then hk^2 is equal to :

(1) 70

(2) 80

(3) 60

(4) 90

Q15 - 2024 (09 Apr Shift 2)

Two vertices of a triangle ABC are $\mathrm{A}(3,-1)$ and $\mathrm{B}(-2,3)$, and its orthocentre is $\mathrm{P}(1,1)$. If the coordinates of the point C are (α,β) and the centre of the of the circle circumscribing the triangle PAB is (h,k), then the value of $(\alpha+\beta)+2(\mathrm{~h}+\mathrm{k})$ equals

(1) 5

(2) 81

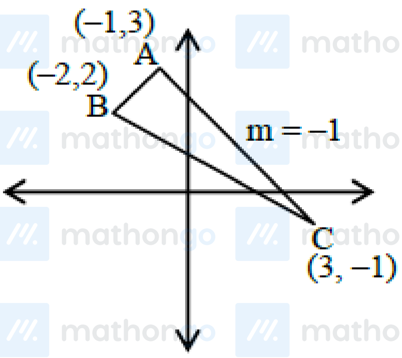
(3) 15

(4) 51

Answer Key

Q1 (3)	**Q2** (14)	**Q3** (2)	**Q4** (2)
Q5 (4)	**Q6** (2)	**Q7** (3)	**Q8** (1)
Q9 (1)	**Q10** (16)	**Q11** (1)	**Q12** (4)
Q13 (1)	**Q14** (1)	**Q15** (1)	

Q1



equation of AC $\rightarrow \mathrm{x}+\mathrm{y}=2$

equation of line parallel to $\mathrm{AC} \mathrm{x}+\mathrm{y}=\mathrm{d}$

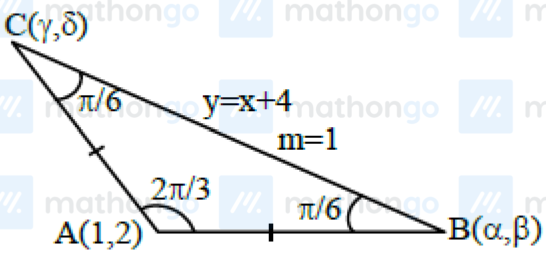
$$\left|\frac{\mathrm{d}-2}{\sqrt{2}}\right|=1$$

$$\mathrm{d}=2-\sqrt{2}$$

eq$^{\mathrm{n}}$ of new required line

$$\mathrm{x}+\mathrm{y}=2-\sqrt{2}$$

Q2



Equation of line passes through point $\mathrm{A}(1,2)$ which makes angle $\frac{\pi}{6}$ from $y=x+4$ is

$$y-2=\frac{1\pm\tan\frac{\pi}{6}}{1\mp\tan\frac{\pi}{6}}(x-1)$$

$$y-2=\frac{\sqrt{3}\pm 1}{\sqrt{3}\mp 1}(x-1)$$

$$\oplus \qquad\qquad \oplus$$

$$\mathrm{y}-2=(2+\sqrt{3})(\mathrm{x}-1) \;\Big|\; \mathrm{y}-2=(2-\sqrt{3})(\mathrm{x}-1)$$

$$\text{solve with } \mathrm{y}=\mathrm{x}+4 \quad \text{solve with } \mathrm{y}=\mathrm{x}+4$$

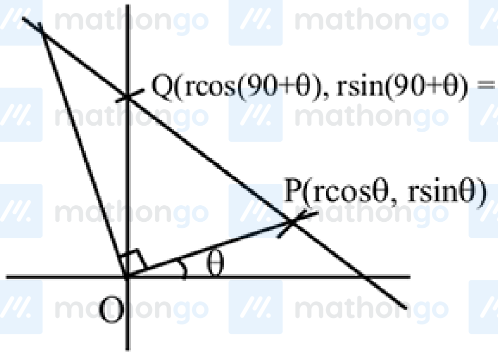
$$x+2=(2+\sqrt{3})x-2-\sqrt{3} \quad x+2=(2-\sqrt{3})x-2+\sqrt{3}$$

$$x=\frac{4+\sqrt{3}}{1+\sqrt{3}} \quad x=\frac{4-\sqrt{3}}{1-\sqrt{3}}$$

$$\alpha^2+\gamma^2=\left(\frac{4+\sqrt{3}}{1+\sqrt{3}}\right)^2+\left(\frac{4-\sqrt{3}}{1-\sqrt{3}}\right)^2$$

$$\alpha^2+\gamma^2=14$$

Q3



$3x + 4y = 12$
$3(r\cos\theta) + 4(r\sin\theta) = 12$
$r(3\cos\theta + 4\sin\theta) = 12 \ldots (1)$
$3(-r\sin\theta) + 4(r\cos\theta) = 12$
$r(-3\sin\theta + 4\cos\theta) = 12 \ldots (2)$

$$\left(\frac{12}{r}\right)^2 + \left(\frac{12}{r}\right)^2 = (3\cos\theta + 4\sin\theta)^2 + (-3\sin\theta + 4\cos\theta)^2$$

$$2\left(\frac{12}{r}\right)^2 = 9 + 16$$

$$\frac{2 \times 144}{r^2} = 25 \Rightarrow 288 = 25r^2$$

$$\Rightarrow \frac{288}{25} = r^2$$

$$\Rightarrow \sqrt{2}\left(\frac{12}{5}\right) = r$$

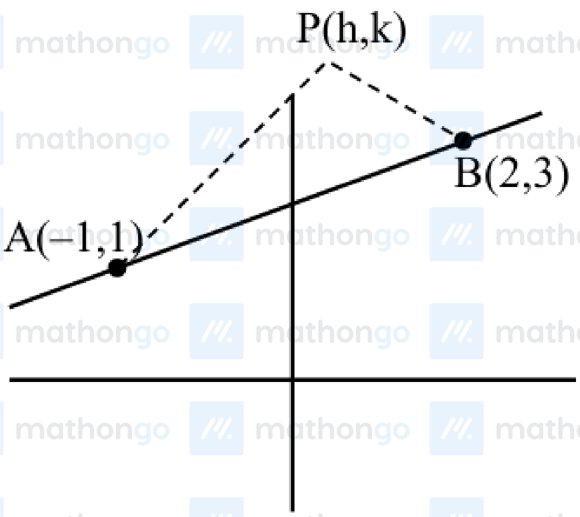
$$\ell = OP^2 + PQ^2 + QO^2$$
$$\ell = r^2 + r^2 \quad + r^2(\cos\theta + \sin\theta)^2 + r^2(\sin\theta + \cos\theta)^2$$
$$= 2r^2 + r^2(1 + \sin 2\theta + 1 - 2\sin 2\theta)$$
$$= 2r^2 + 2r^2$$
$$= 4r^2$$
$$= 4\left(\frac{288}{25}\right) = \frac{1152}{25} = 46.08$$
$$[\ell] = 46$$

Q4

$A(1,-1,2)$
$B(5,7,-6)$
$C(3,4,-10)$
$D(-1,-4,-2)$

$$\text{Area }=\frac{1}{2}|\overrightarrow{AC}\times\overrightarrow{BD}|=\frac{1}{2}|(2\hat{i}+5\hat{j}-12\hat{k})\times(6\hat{i}+11\hat{j}-4\hat{k})|$$
$$=\frac{1}{2}|112\hat{i}-64\hat{j}-8\hat{k}|$$
$$=4|14\hat{i}-8\hat{j}-\hat{k}|$$
$$=4\sqrt{196+64+1}$$
$$=4\sqrt{261}$$
$$=12\sqrt{29}$$

Q5



$$\frac{1}{2}\begin{vmatrix} h & k & 1 \\ -1 & 1 & 1 \\ 2 & 3 & 1 \end{vmatrix}=10$$
$$-2x+3y=25$$
$$-\frac{6}{5}x+\frac{9}{5}y=15$$
$$a=-\frac{6}{5},\ b=\frac{9}{5}$$
$$5a=-6, 2\,b=\frac{18}{5}$$

Q6

equation of line is

$y+9=m(x-4)$

$\therefore \quad A=\left(\frac{9+4m}{m}, 0\right)$

$B=(0,-9-4m)$

$\therefore \quad OA+OB=\frac{9+4m}{m}+9+4m$

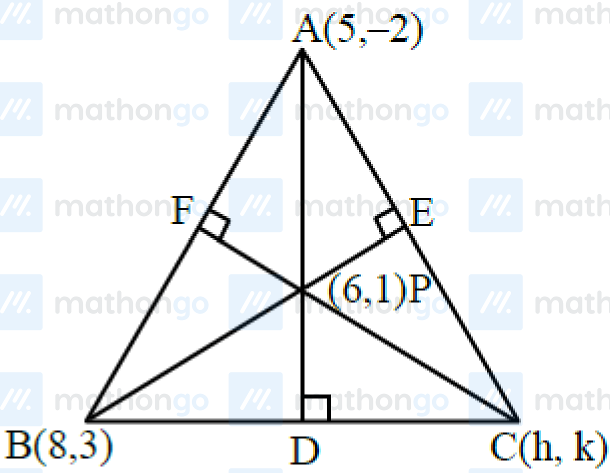
$\because \mathrm{m}>0$

$=13+\frac{9}{\mathrm{m}}+4\mathrm{~m}$

$\because \frac{4\mathrm{~m}+\frac{9}{\mathrm{m}}}{2} \geq \sqrt{36} \Rightarrow 4\mathrm{~m}+\frac{9}{\mathrm{m}} \geq 12$

$\therefore \mathrm{OA}+\mathrm{OB} \geq 25$

Q7



Slope of $AD=3$

Slope of $\mathrm{BC}=-\frac{1}{3}$

equation of $\mathrm{BC}=3\mathrm{y}+\mathrm{x}-17=0$

slope of $\mathrm{BE}=1$

Slope of $\mathrm{AC}=-1$

equation of AC is $\mathrm{x}+\mathrm{y}-3=0$

point C is $(-4,7)$

Q8

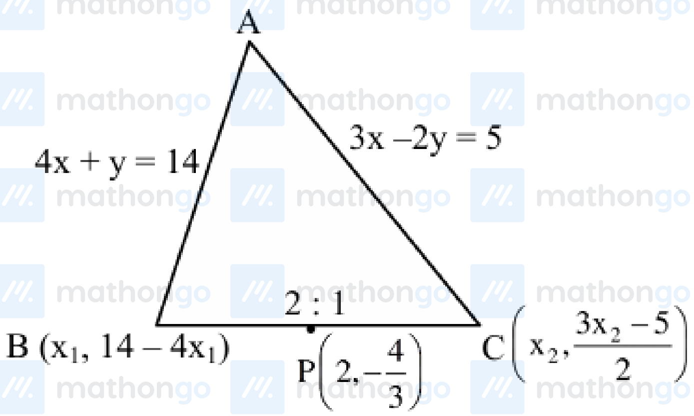
let $P(x, y)$

$$\frac{(x-2)^2+(y-1)^2}{(x-1)^2+(y-3)^2}=\frac{25}{16}$$

$$9x^2+9y^2+14x-118y+170=0$$

$$a^2+2b+3c+4d+e$$
$$=81+18+0+56-118$$
$$=155-118$$
$$=37$$

Q9



$$\frac{2x_2+x_1}{3}=2,\ \frac{2\left(\frac{3x_2-5}{2}\right)+(14-4x_1)}{3}=\frac{-4}{3}$$

$$2x_2+x_1=6,\ 3x_2-4x_1=-13$$
$$x_2=1,\ x_1=4$$

So, $C(1,-1), B(4,-2)$

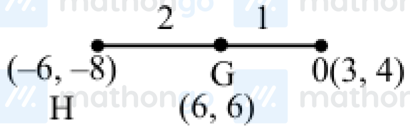
$m=\frac{-1}{3}$

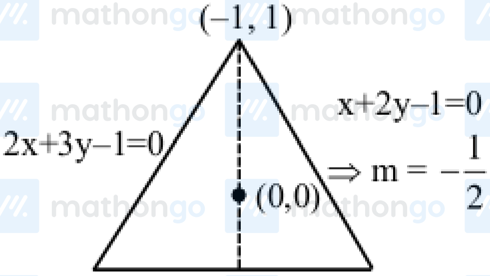
Equation of BC : $y+1=\frac{-1}{3}(x-1)$

$$3y+3=-x+1$$
$$x+3y+2=0$$

Q10

$$2x+3y-1=0$$
$$x+2y-1=0$$
$$ax+by-1=0$$





$\left(\frac{6-6}{3}, \frac{8-8}{3}\right)$
$= (0,0)$
$ax + by - 1 = 0$
$\left(\frac{1-0}{-1-0}\right)\left(\frac{-a}{b}\right) = -1$
$\Rightarrow -\mathrm{a} = \mathrm{b}$
$\Rightarrow \quad \mathrm{ax} - \mathrm{ay} - 1 = 0$
$ax - a\left(1 - \frac{2x}{3}\right) - 1$
$x\left(a + \frac{2a}{3}\right) = \frac{a}{3}$
$\mathrm{x} = \frac{\mathrm{a}+3}{5\mathrm{a}}$
$2\left(\frac{a+3}{5a}\right) + 3y - 1 = 0$
$y = \frac{1 - \frac{2a+6}{5a}}{3} = \frac{3a-6}{3 \times 5a}$
$y = \frac{a-2}{5a}$
$\frac{\left(\frac{a-2}{5a}\right)}{\left(\frac{a+3}{5a}\right)} = 2 \Rightarrow a - 2 = 2a + 6$
$a = -8$
$\mathrm{b} = 8$
$-8\mathrm{x} + 8\mathrm{y} - 1 = 0$
$|a-b| = 16$

Q11

Image of point $(-4, 5)$

$\frac{x-x_1}{a} = \frac{y-y_1}{b} = -2\left(\frac{ax_1+by_1+c}{a^2+b^2}\right)$

Line: $x + 2y - 2 = 0$

$\frac{x+4}{1} = \frac{y-5}{2} = -2\left(\frac{-4+10-2}{1^2+2^2}\right)$

$= \frac{-8}{5}$

$x = -4 - \frac{8}{5} = -\frac{28}{5}$

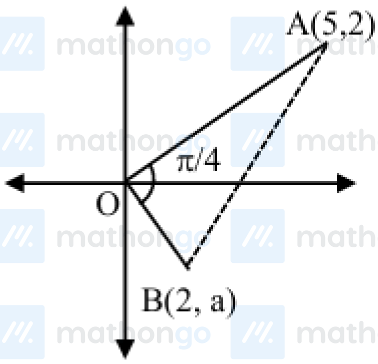
$y = -\frac{16}{5} + 5 = \frac{9}{5}$

Point lies on circle $(x+4)^2 + (y-3)^2 = r^2$

$\frac{64}{25} + \left(\frac{9}{5} - 3\right)^2 = r^2$

$\frac{100}{25} = r^2, r = 2$

Q12



$m_{OA} = \frac{2}{5}$

$m_{OB} = \frac{a}{2}$

$4 - 5a = \pm(10 + 2a)$

$4 - 5a = 10 + 2a$ $\quad$ $4 - 5a = -10 - 2a$

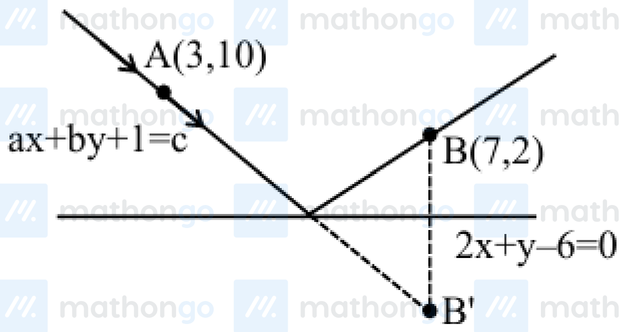
$\Rightarrow 7a + 6 = 0$ $\quad$ $3a = 14$

$\Rightarrow a = -\frac{6}{7}$ $\quad$ $a = +\frac{14}{3}$

$\tan\frac{\pi}{4} = \left|\frac{\frac{2}{5} - \frac{a}{2}}{}\right|$ $-\frac{6}{7} \times \frac{14}{3} = -4$

$1 = \left|\frac{4 - 5a}{10 + 2a}\right|$

Q13



For B' $\frac{x-7}{2}=\frac{y-2}{1}=-2\left(\frac{14+2-6}{5}\right)$

$\frac{x-7}{2}=\frac{y-2}{1}=-4$

$x=-1 \quad y=-2 \quad B'(-1,-2)$

incident ray AB'

$M_{\mathrm{AB}'}=3$

$y+2=3(x+1)$

$3x-y+1=0$

$a=3b=-1$

$a^2+b^2+3ab=9+1-9=1$

Q14

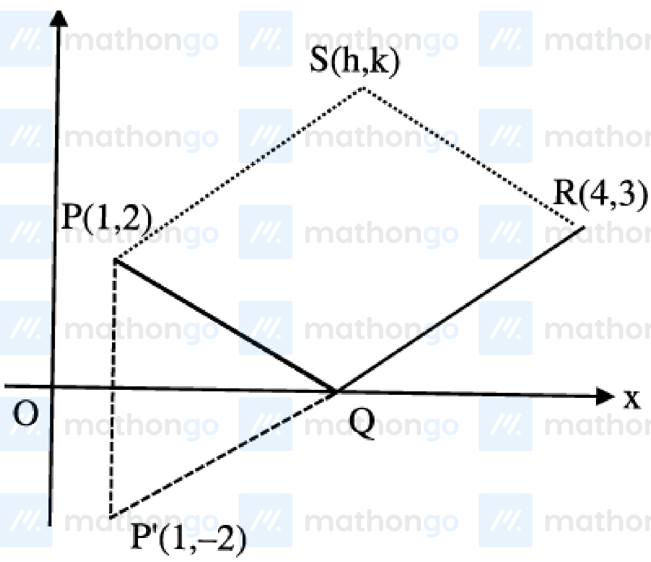


Image of P wrt x-axis will be $P'(1,-2)$ equation of line joining $P'R$ will be

$y-3=\frac{5}{3}(x-4)$

Above line will meet x-axis at Q where

$y=0 \Rightarrow x=\frac{11}{5}$

$\therefore Q\left(\frac{11}{5}, 0\right)$

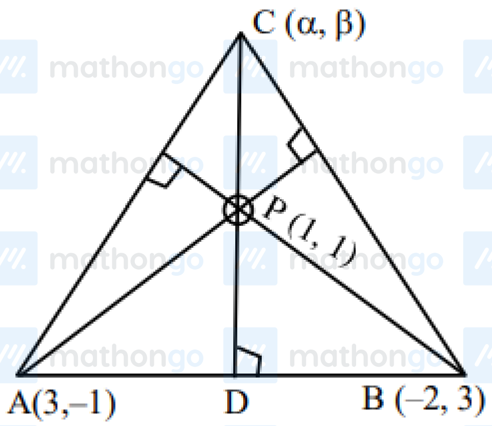
$\because$ PQRS is parallelogram so their diagonals will bisects each other

$\Rightarrow \frac{4+1}{2}=\frac{\frac{11}{5}+h}{2} \& \frac{2+3}{2}=\frac{k+0}{2}$

$\Rightarrow h=\frac{14}{5} \& k=5$

$\therefore hk^2=\frac{14}{5} \times 5^2=70$

Q15



$M_{AB}=\frac{4}{-5} \Rightarrow M_{DP}=\frac{5}{4}$

Equation of PC is $y-1=\frac{5}{4}(x-1) \ldots(1)$

$M_{AP}=\frac{2}{-2}=-1 \Rightarrow M_{BC}=+1$

Equation of BC is $y-3=(x+2) \ldots(2)$

On solving (1) and (2)

$x+4=\frac{5}{4}(x-1) \Rightarrow 4x+16=5x-5 \Rightarrow \alpha=21$

$\Rightarrow \beta=y=x+5=26$

$\alpha+\beta=47$

Equation of $\perp$ bisector of AP

$y-0=(x-2) \ldots(3)$

Equation of $\perp$ bisector of AB

$y-1=\frac{5}{4}\left(x-\frac{1}{2}\right) \ldots(4)$

On solving (3) \& (4)

$(\mathrm{x}-3) 4=5 \mathrm{x}-\frac{5}{2}$

$\mathrm{x}=\frac{-19}{2}=\mathrm{h}$

$\mathrm{y}=\frac{-23}{2}=\mathrm{k}$

$\Rightarrow 2(\mathrm{~h}+\mathrm{k})=-42$